

Class/ Index Number	Centre Number/ 'O' Level Index Number	Name SOLUTIONS
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新加坡海星中学
MARIS STELLA HIGH SCHOOL
PRELIMINARY EXAMINATION
SECONDARY FOUR

ADDITIONAL MATHEMATICS

4049/1

Paper 1

20 August 2026

Candidates answer on the Question Paper.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

For Examiners' Use

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
/ 5	/ 6	/ 7	/ 5	/ 6	/ 7	/ 8	/ 4	/ 7
Q10	Q11	Q12	Q13	SUBTOTAL			90	
/ 9	/ 8	/ 10	/ 8	Statement				
				Presentation				
				Units				
				Rounding Off				

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

- 1** A sum of \$500 is invested in an account that pays compound interest at a rate of 6% per year. The interest is compounded annually. The investment is withdrawn in the year when the total **interest** exceeds \$300. Find the least number of years the money must be invested before it is withdrawn. [5]

[It is given that for compound interest, total amount = $P\left(1 + \frac{r}{100}\right)^n$]

$$A = P\left(1 + \frac{r}{100}\right)^n$$

$$500(1.06)^n > 500 + 300$$

$$500(1.06)^n > 800$$

$$(1.06)^n > \frac{8}{5}$$

$$n \ln(1.06) > \ln\left(\frac{8}{5}\right)$$

$$n > 8.06$$

$$\therefore n = 9$$

- 2 Find the set of values of the constant m for which the curve $y = mx^2 + 4x + m + 3$ is completely above the line $y = 4x + 12$. [6]

Since curve is above the line, $m > 0$.

Sub line into curve:

$$mx^2 + 4x + m + 3 = 4x + 12$$

$$mx^2 + m - 9 = 0$$

$$b^2 - 4ac < 0$$

$$0 - 4m(m - 9) < 0$$

$$-4m^2 + 36m < 0$$

$$m^2 - 9m > 0$$

$$m(m - 9) > 0$$

$$\therefore m < 0 \text{ or } m > 9$$

Since the curve is above the line,

$$\therefore m > 9$$

- 3 (a) Differentiate $y = \frac{\ln x}{x}$ with respect to x . [3]

$$y = \frac{\ln x}{x}$$

$$\frac{dy}{dx} = \frac{x\left(\frac{1}{x}\right) - \ln x(1)}{x^2}$$

$$= \frac{1 - \ln x}{x^2}$$

- (b) Hence find $\int_1^e -\frac{\ln x}{x^2} dx$. [4]

$$\int \frac{1 - \ln x}{x^2} dx = \frac{\ln x}{x} + c$$

$$\int \frac{1}{x^2} - \frac{\ln x}{x^2} dx = \frac{\ln x}{x} + c$$

$$\int -\frac{\ln x}{x^2} dx = \frac{\ln x}{x} - \int \frac{1}{x^2} dx + c$$

$$\int_1^e -\frac{\ln x}{x^2} dx = \left[\frac{\ln x}{x} \right]_1^e - \int_1^e \frac{1}{x^2} dx$$

$$\int_1^e -\frac{\ln x}{x^2} dx = \left(\frac{\ln e}{e} \right) - \left(\frac{\ln 1}{1} \right) - \left[-\frac{1}{x} \right]_1^e$$

$$\therefore \int_1^e -\frac{\ln x}{x^2} dx = \frac{1}{e} - \left[\left(-\frac{1}{e} \right) - \left(-\frac{1}{1} \right) \right] = -0.264 \text{ (3 sf)}$$

4 Express $\frac{2x^2 - 7x - 5}{x^2 - 2x - 3}$ in partial fractions.

[5]

Using long division, $\frac{2x^2 - 7x - 5}{x^2 - 2x - 3} = 2 + \frac{1 - 3x}{x^2 - 2x - 3}$

$$\frac{1 - 3x}{x^2 - 2x - 3} = \frac{A}{x - 3} + \frac{B}{x + 1}$$

$$1 - 3x = A(x + 1) + B(x - 3)$$

When $x = -1$,

$$4 = -4B \Rightarrow B = -1$$

When $x = 3$,

$$-8 = 4A \Rightarrow A = -2$$

$$\therefore \frac{2x^2 - 7x - 5}{x^2 - 2x - 3} = 2 - \frac{2}{x - 3} - \frac{1}{x + 1}$$

$$\frac{2x^2 - 7x - 5}{x^2 - 2x - 3} = 2 - \frac{2}{x - 3} - \frac{1}{x + 1}$$

- 5 (a) Show that the x -coordinate of the stationary point on the curve $y = \frac{1}{3}x^3 - 3x^{\frac{2}{3}} + 4$ is $2^{\frac{3}{7}}$. [4]

$$y = \frac{1}{3}x^3 - 3x^{\frac{2}{3}} + 4$$

$$\text{Let } \frac{dy}{dx} = x^2 - 2x^{-\frac{1}{3}} = 0$$

$$x^2 = \frac{2}{\sqrt[3]{x}}$$

$$x^{\frac{7}{3}} = 2$$

$$\therefore x = 2^{\frac{3}{7}}$$

- (b) Determine the nature of this stationary point. [2]

Method 1

x	$\left(2^{\frac{3}{7}}\right)^{-}$ (e.g. 1.34)	$2^{\frac{3}{7}}$	$\left(2^{\frac{3}{7}}\right)^{+}$ (e.g. 1.35)
$\frac{dy}{dx}$	$-0.0185 < 0$	0	$2.72 > 0$
Slope	\	-	/

Method 2

$$\frac{dy}{dx} = x^2 - 2x^{-\frac{1}{3}}$$

$$\frac{d^2y}{dx^2} = 2x + \frac{2}{3}x^{-\frac{4}{3}}$$

$$\text{When } x = 2^{\frac{3}{7}}, \frac{d^2y}{dx^2} = 3.14 > 0 \Rightarrow \text{min pt.}$$

- 6 The force, F Newtons, acting on a moving vehicle depends on its speed, s m/s and can be modelled by the formula $F = As^k$, where A and k are constants. The table below shows corresponding values of s and F .

s	2	4	6	8	10
F	6	22	50	80	150

- (a) Explain how a straight line graph can be drawn to represent the formula and draw the graph on the grid on page 9. [4]

$$F = As^k$$

$$\ln F = \ln(As^k)$$

$$\ln F = \ln A + k \ln s$$

Plot $\ln F$ against $\ln s$

$\ln s$	0.69	1.39	1.79	2.08	2.30
$\ln F$	1.79	3.09	3.91	4.38	5.01

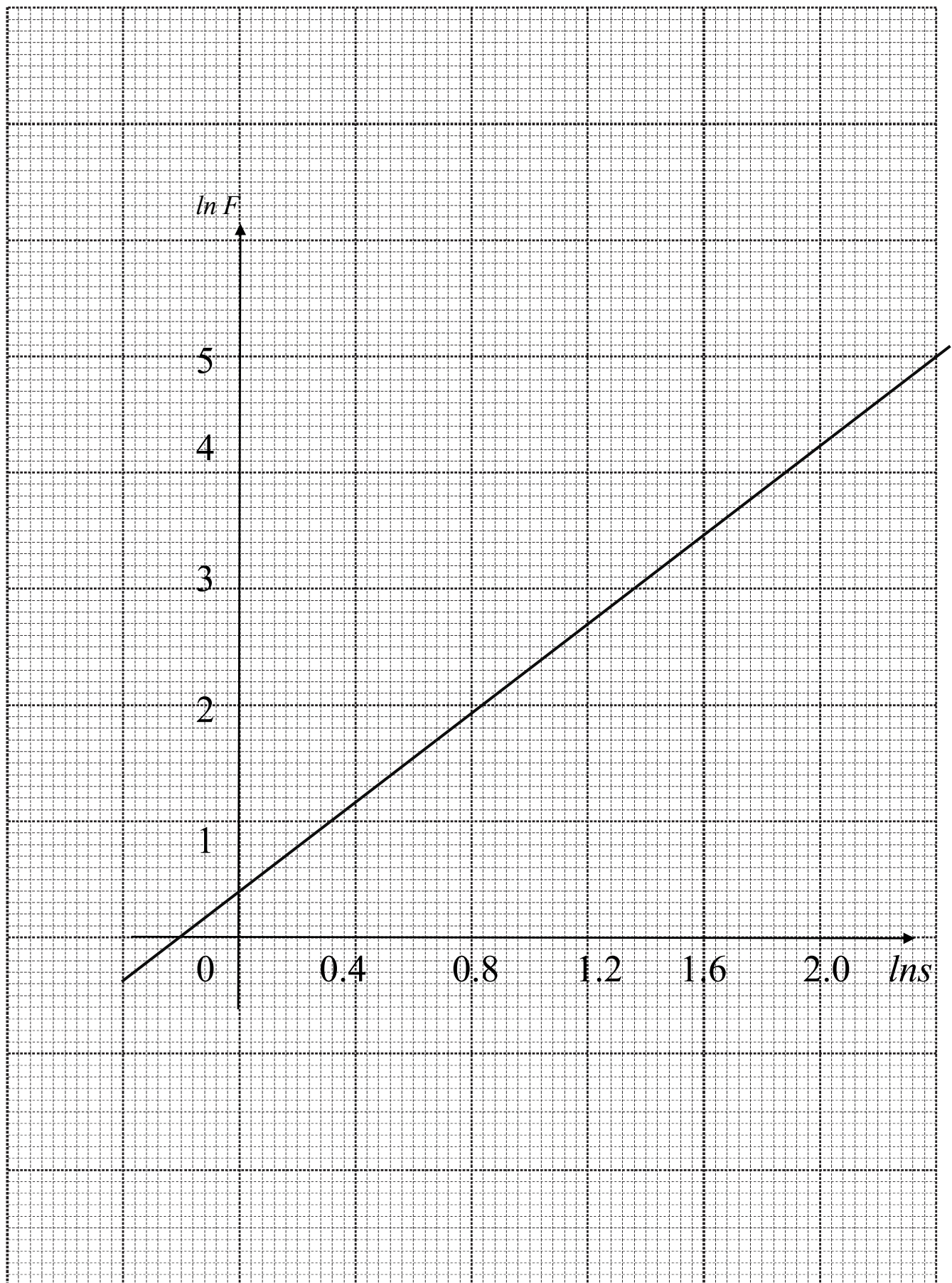
OR

$\lg s$	0.30	0.60	0.78	0.90	1
$\lg F$	0.78	1.34	1.70	1.90	2.18

- (b) Use your graph to estimate the values of A and k . [3]

$$k = 1.90 \text{ (accept } \pm 0.15 \text{)}$$

$$A = e^{0.4} = 1.49 \text{ (3 sf) (accept 1.35 to 1.65)}$$



- 7 (a) Find, in ascending powers of x , the first four terms in the expansion of $\left(\frac{1}{3} - bx\right)^6$, where b is a constant. Simplify the coefficients in your expansion. [4]

$$\begin{aligned}\left(\frac{1}{3} - bx\right)^6 &= \left(\frac{1}{3}\right)^6 + \binom{6}{1}\left(\frac{1}{3}\right)^5(-bx)^1 + \binom{6}{2}\left(\frac{1}{3}\right)^4(-bx)^2 + \binom{6}{3}\left(\frac{1}{3}\right)^3(-bx)^3 + \dots \\ &= \frac{1}{729} - \frac{2}{81}bx + \frac{5}{27}b^2x^2 - \frac{20}{27}b^3x^3 + \dots\end{aligned}$$

- (b) Given that the expansion of $\left(1 + \frac{3}{x} + \frac{3}{x^2}\right)\left(\frac{1}{3} - bx\right)^6$ has no constant term, find the possible values of b . [4]

$$\begin{aligned}&\left(1 + \frac{3}{x} + \frac{3}{x^2}\right)\left(\frac{1}{3} - bx\right)^6 \\ &= \left(1 + \frac{3}{x} + \frac{3}{x^2}\right)\left(\frac{1}{729} - \frac{2}{81}bx + \frac{5}{27}b^2x^2 - \frac{20}{27}b^3x^3 + \dots\right)\end{aligned}$$

$$\text{Constant term} = \frac{1}{729} - \frac{2}{27}b + \frac{5}{9}b^2$$

$$\frac{1}{729} - \frac{2}{27}b + \frac{5}{9}b^2 = 0$$

$$405b^2 - 54b + 1 = 0$$

$$(9b - 1)(45b - 1) = 0$$

$$\therefore b = \frac{1}{9} \text{ or } b = \frac{1}{45}$$

- 8 A circle has equation $x^2 + y^2 - 6x + 2y = a$, where a is a constant. Find the set of values of a for which there are two real roots when the line $y = 2x - 5$ intersects the circle. [4]

$$x^2 + y^2 - 6x + 2y = a$$

$$x^2 + (2x - 5)^2 - 6x + 2(2x - 5) = a$$

$$x^2 + 4x^2 - 20x + 25 - 6x + 4x - 10 - a = 0$$

$$5x^2 - 22x + 15 - a = 0$$

$$b^2 - 4ac \geq 0$$

$$22^2 - 4(5)(15 - a) \geq 0$$

$$484 - 300 + 20a \geq 0$$

$$20a \geq -184$$

$$\therefore a \geq -9.2$$

9 (a) Prove that $\frac{1+\cos\theta}{\sin\theta} - \frac{\sin\theta}{1+\cos\theta} = 2\cot\theta$. [4]

$$\begin{aligned}
 \text{LHS} &= \frac{1+\cos\theta}{\sin\theta} - \frac{\sin\theta}{1+\cos\theta} \\
 &= \frac{(1+\cos\theta)^2}{\sin\theta(1+\cos\theta)} - \frac{\sin^2\theta}{\sin\theta(1+\cos\theta)} \\
 &= \frac{(1+\cos\theta)^2 - \sin^2\theta}{\sin\theta(1+\cos\theta)} \\
 &= \frac{1+2\cos\theta+\cos^2\theta - \sin^2\theta}{\sin\theta(1+\cos\theta)} \\
 &= \frac{1+2\cos\theta+\cos^2\theta - (1-\cos^2\theta)}{\sin\theta(1+\cos\theta)} \\
 &= \frac{2\cos\theta+2\cos^2\theta}{\sin\theta(1+\cos\theta)} \\
 &= \frac{2\cos\theta(1+\cos\theta)}{\sin\theta(1+\cos\theta)} \\
 &= 2\cot\theta
 \end{aligned}$$

(b) Hence solve $\frac{1+\cos\theta}{\sin\theta} - \frac{\sin\theta}{1+\cos\theta} = 1$ for $0^\circ < \theta < 360^\circ$. [3]

$$\frac{1+\cos\theta}{\sin\theta} - \frac{\sin\theta}{1+\cos\theta} = 1$$

$$2\cot\theta = 1$$

$$\tan\theta = 2$$

$$\alpha = 63.435^\circ$$

θ lies in 1st and 3rd quadrants.

$$\theta = 63.435^\circ \text{ or } 243.43^\circ$$

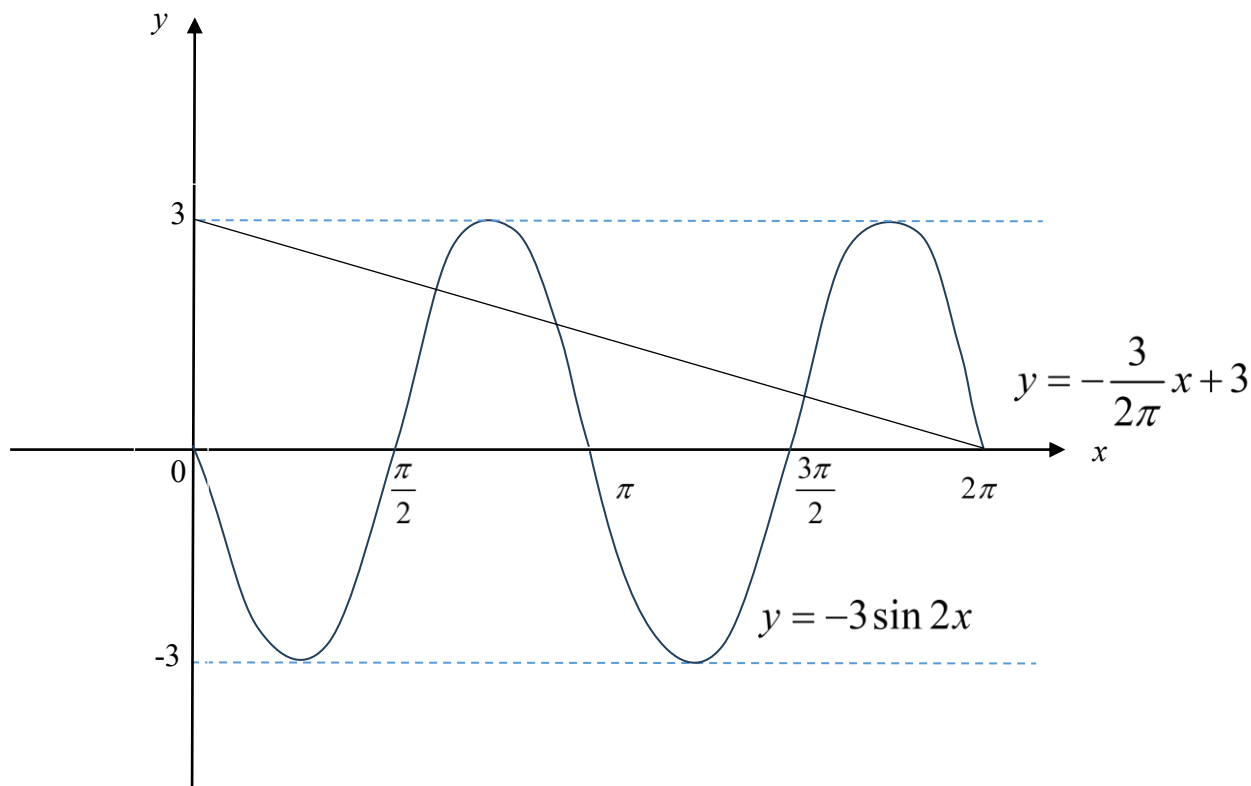
$$\theta = 63.4^\circ \text{ or } 243.4^\circ \text{ (1 dp)}$$

- 10 (a) State the amplitude and period of $-3\sin 2x$. [2]

3,

180° or π

- (b) Sketch the graph of $y = -3\sin 2x$ for $0 \leq x \leq 2\pi$. [3]



- (c) The equation $-3\sin 2x = k$ has 2 solutions. State a possible value of k . [1]

3 or -3

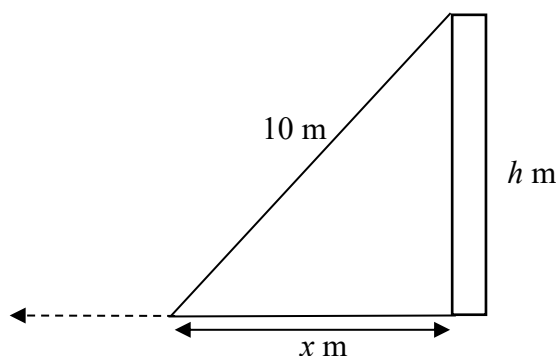
- (d) By drawing a suitable straight line on your sketch, determine the number of solutions of the equation $-\frac{1}{2\pi}x + 1 = -\sin 2x$. [3]

$$-\frac{1}{2\pi}x + 1 = -\sin 2x$$

$$-\frac{3}{2\pi}x + 3 = -3\sin 2x \Rightarrow \text{To sketch } y = -\frac{3}{2\pi}x + 3.$$

Sketch straight line graph on sketch
4 solutions.

- 11** A ladder of fixed length 10 m leans against a vertical wall. The base of the ladder slides away from the wall along the horizontal ground. At time t seconds, the base of the ladder is x m from the wall, and the height the ladder reaches up the wall is h m. The base of the ladder is sliding away from the wall at a constant rate of 0.5 m per second.



- (a) Show that h is decreasing at a constant rate of 0.375 m per second when $x = 6$. [5]

Given $\frac{dx}{dt} = 0.5$, find $\frac{dh}{dt}$ when $x = 6$ and we need $\frac{dh}{dx}$.

$$h^2 + x^2 = 10^2$$

$$h^2 = 100 - x^2$$

$$h = \sqrt{100 - x^2} \text{ since } x > 0$$

When $x = 6$, $h = 8$.

$$\frac{dh}{dx} = \frac{1}{2}(100 - x^2)^{-\frac{1}{2}}(-2x) = \frac{-x}{\sqrt{100 - x^2}}$$

$$\frac{dh}{dt} = \frac{dh}{dx} \times \frac{dx}{dt}$$

$$\frac{dh}{dt} = \frac{-x}{\sqrt{100 - x^2}} \times (0.5)$$

$$\frac{dh}{dt} = \frac{-6}{8}(0.5) = -0.375 \text{ m/sec (shown)}$$

- (b) Find the rate of change of the area of the triangle formed by the ladder, the wall and the ground with respect to t when $x = 6$. [3]

$$A = \frac{1}{2} xh = \frac{1}{2} x\sqrt{100-x^2}$$

$$\begin{aligned}\frac{dA}{dx} &= \frac{1}{2} x \left(\frac{1}{2} \right) (100-x^2)^{-\frac{1}{2}} (-2x) + \frac{1}{2} \sqrt{100-x^2} \\ &= -\frac{1}{2} x^2 (100-x^2)^{-\frac{1}{2}} + \frac{1}{2} \sqrt{100-x^2}\end{aligned}$$

$$\text{When } x = 6, \frac{dA}{dx} = 1.75$$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt} = 1.75 \times (0.5) = 0.875 \text{ m}^2 / \text{sec}$$

12 A curve has equation $y = 5 \cos 3x - 2 \sin ax$, where a is a positive constant.

- (a) Find, in terms of a , the equations of the tangent and normal to the curve at the point R where $x = 0$. [6]

$$y = 5 \cos 3x - 2 \sin ax$$

$$\frac{dy}{dx} = -15 \sin 3x - 2a \cos ax$$

$$\text{When } x = 0, \frac{dy}{dx} = -2a, y = 5.$$

$$\text{Gradient of normal} = \frac{1}{2a}$$

Equation of normal:

$$y - 5 = \frac{1}{2a}(x - 0)$$

$$y = \frac{1}{2a}x + 5$$

Equation of tangent:

$$y - 5 = -2a(x - 0)$$

$$y = -2ax + 5$$

- (b) The tangent cuts the x -axis at the point S and the normal cuts the y -axis at the point T . Given that length ST is 6 units, find the value of a . [4]

$$\text{When } y = 0, x = \frac{5}{2a} \Rightarrow S\left(\frac{5}{2a}, 0\right)$$

$$\text{When } x = 0, y = 5 \Rightarrow T(0, 5)$$

$$\therefore \sqrt{\left(\frac{5}{2a} - 0\right)^2 + (0 - 5)^2} = 6$$

$$\sqrt{\frac{25}{4a^2} + 25} = 6$$

$$\frac{25}{4a^2} + 25 = 36$$

$$\frac{25}{4a^2} = 11$$

$$a = 0.754 \text{ or } \frac{5\sqrt{11}}{22} \text{ (since } a > 0)$$

- 13** A telecommunications company is setting up a circular coverage zone. Two engineers are at points $A(5,3)$ and $B(-1,3)$ on the perimeter of the circle. The centre of the coverage zone lies on the line $y = 2x - 5$.

(a) Find the equation of the circle in the form $x^2 + y^2 + 2gx + 2fy + c = 0$. [6]

$$\text{Midpoint of } AB = (2, 3)$$

$$\text{Gradient of } AB = 0$$

$$\therefore \text{Equation of perpendicular bisector: } x = 2 \quad \text{--- (1)}$$

$$y = 2x - 5 \quad \text{--- (2)}$$

$$\therefore \text{Centre of the circle} = (2, -1)$$

$$\text{Radius} = \sqrt{(2-5)^2 + (-1-3)^2} = 5$$

$$\therefore \text{Equation of circle: } (x-2)^2 + (y+1)^2 = 25$$

$$\Rightarrow x^2 + y^2 - 4x + 2y - 20 = 0$$

- (b) A signal booster is placed at the point $(2, m)$ inside the coverage zone. Find the range of values of m . [2]

$$\sqrt{(2-2)^2 + (-1-m)^2} < 5$$

$$(-1-m)^2 < 25$$

$$m^2 + 2m + 1 - 25 < 0$$

$$(m+6)(m-4) < 0$$

$$\therefore -6 < m < 4$$